ANALYTICAL SQL PROJECT

Business Insights from Sakila DVD Rentals

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Project Objectives

Customer Analytics: Profiling participation and identifying high-value spending patterns.

Revenue Optimization: Quantifying totals and identifying the most profitable genres.

Operational Intelligence: Comparing store performance and seasonality to optimize inventory.

The Tooling Stack

MySQL
The core engine used for managing and querying the relational database.

SQL
Advanced DQL (Data Query Language) used for multi-way joins and aggregations.

Workbench
Graphical interface used for visual schema design and query optimization.

Market Scale & Volume

599

Active Customers

A substantial base of repeat rental activity provides a strong dataset for analysis.

~16K

Total Rentals

Strong customer engagement across the entire film catalog.

Average Spend Per Customer: $112.53

Identifying High-Value VIPs

Spotlight: Karl Seal

Identification of VIP customers allows for highly targeted marketing and loyalty programs.

- Top Spender in the entire system
- Total Lifetime Spent: **$221.55**
- Consistent high-frequency rental behavior

Revenue by Movie Category

Flagship Titles & Inventory

Most Rented Film

"Bucket Brotherhood"

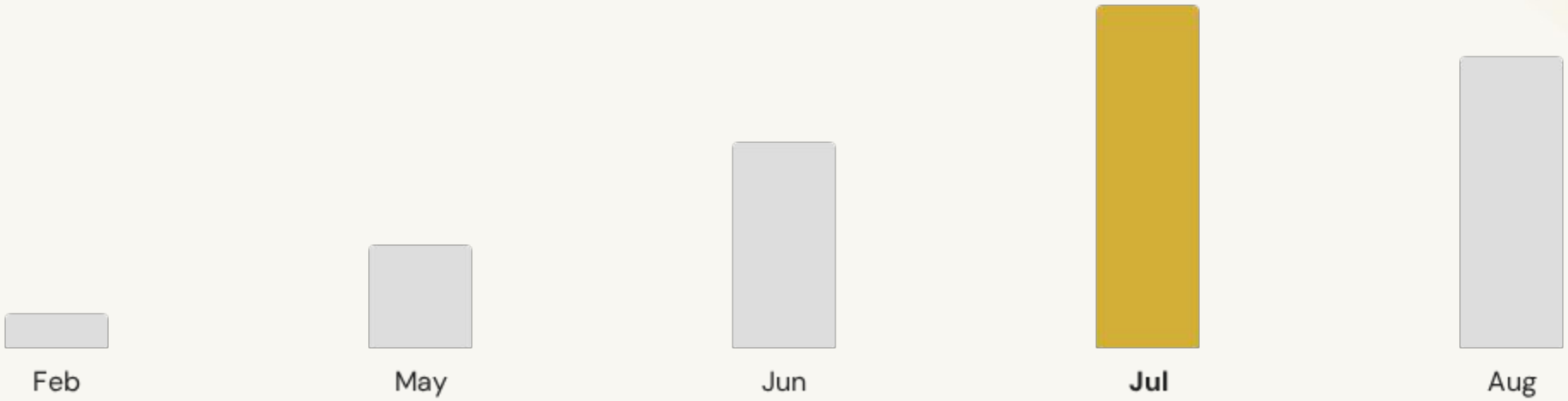
With 34 rentals, this title represents our most consistent asset.

- "Rocketeer Mother" (33 rentals)
- "Scalawag Duck" (32 rentals)
- "Ridgemont Submarine" (32 rentals)

Monthly Rental Trends

The Peak Performance Month: **July**

Rental demand exhibits clear seasonal patterns rather than a uniform distribution.



Store Performance Comparison

Store 1

$36,998

Total Revenue Captured

Customers: 326

Store 2

$30,409

Total Revenue Captured

Customers: 273

Store 1 exhibits a **21% higher revenue** yield than Store 2.

The Star Power Effect

Actor Demand Analysis

Does actor presence influence rental choices? Data suggests a direct correlation.

Dina Degeneres
Appears in 42 films with 753 cumulative rentals.

Matthew Carrey
Associated with 678 total rentals.

- Popular actors can be leveraged in marketing campaigns.
- Films featuring these actors warrant higher inventory counts.
- Use "Star Power" to drive recommendation engine logic.

Executive Summary

Consolidated Insights: SQL analysis revealed a strong $67k revenue base with high seasonality peaks in July.

Strategic Action: Target the Sports category for acquisition and Karl Seal-type VIPs for retention.

Scale: Managing 16k transactions across 2 stores shows efficient operational throughput.